

Newline-Only Transcription Alignment For Historical And Modern Documents With Vision-Language Models

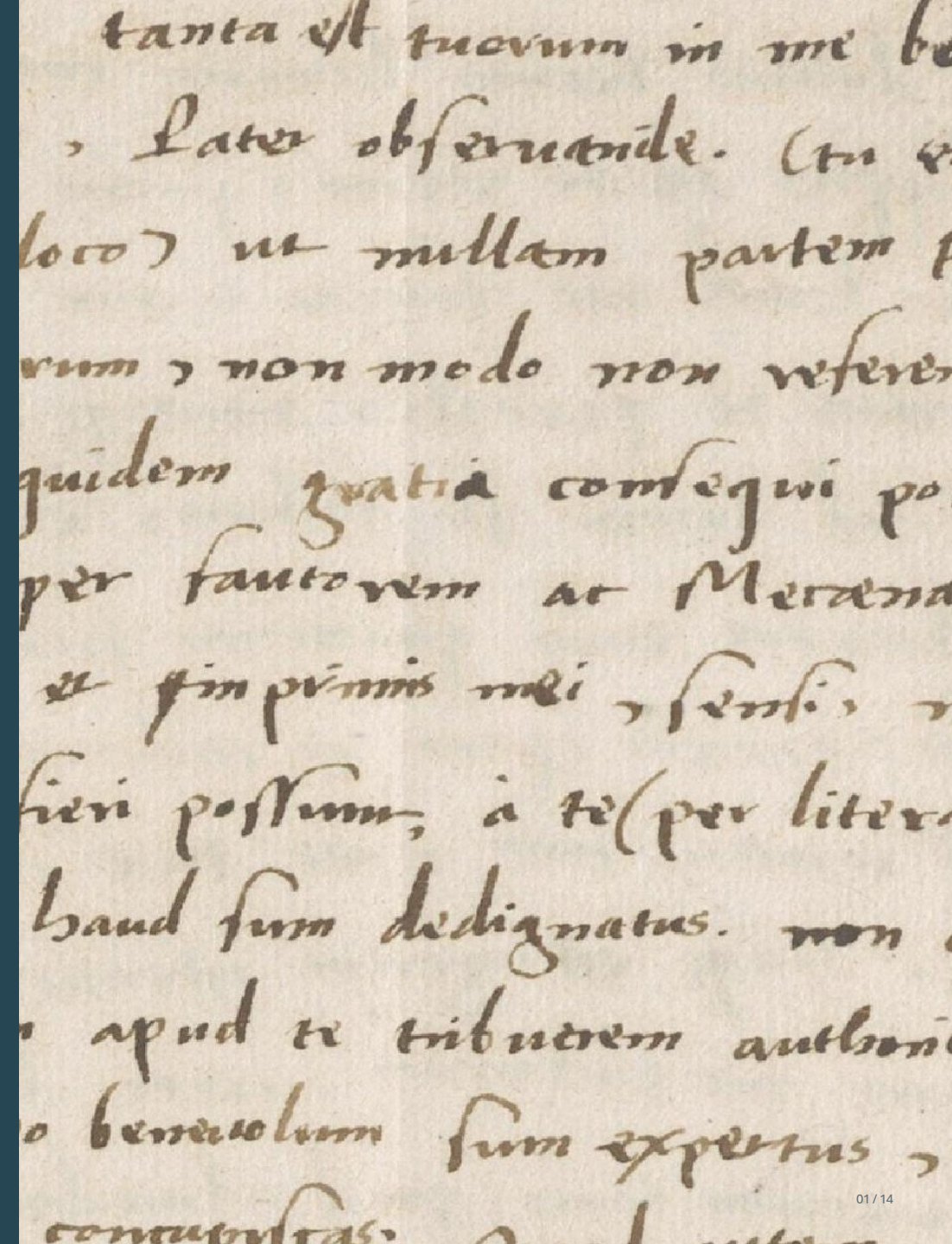
Master's Thesis Defense.

Dennis Berger

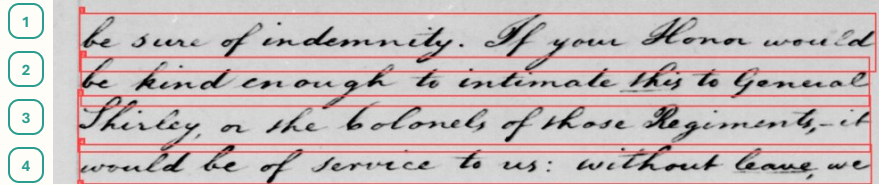
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June 16, 2026



page image crop



page-level transcription

be sure of idemnity. If your Honor would be kind enough to intimate this to General Shirley, or the Colonels of those Regiments,- it would be of service to us: without leave we



aligned line transcription

- 1 be sure of idemnity. If your Honor would
- 2 be kind enough to intimate this to General
- 3 Shirley, or the Colonels of those Regiments,- it
- 4 would be of service to us: without leave we

Line Evidence Enables Inspection

Line boundaries connect transcription back to the visual page; in the document pipeline, newline-only alignment links available structure to the fixed transcription.

Why VLMs

- Avoid training a task-specific aligner for every evidence setting
- Combine page image, line crops, OCR/HTR hints, and fixed transcription
- Use visual layout and textual context together

Challenges

- Generative output may omit, normalize, or add text
- Copied text is not trusted as final content
- Transcription alignment needs post-processing

Evaluate input combinations + control setups, not just models.

RQ1

Which Input And Method Variants Improve Newline-Boundary Accuracy For Fixed Transcriptions?

Compare input and method variants under fixed transcription

RQ2

How Can Generated Outputs Be Constrained So The Fixed Transcription Is Preserved?

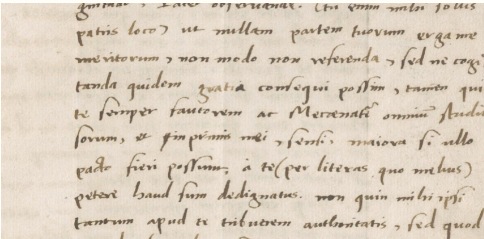
Constrain outputs through repair, recovery, fallback, projection

RQ3

How Does The Highest-Scoring Evaluated VLM Setup Compare With The Deterministic NNTP Reference?

Compare highest-scoring evaluated VLM setup with deterministic NNTP reference

Contribution: Newline-only protocol/benchmark | input/method variants | repair/recovery/fallback/projection | projection preserves fixed transcription | deterministic NNTP comparison

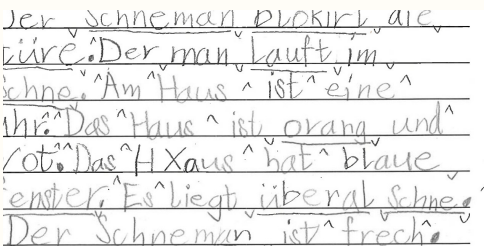


Bullinger

Latin / Premodern German / Multilingual

handwritten

59 samples
100 pages
2,535 lines
133,883 chars

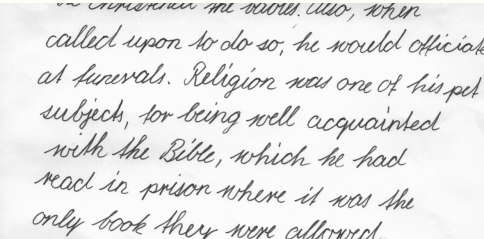


Children

German Child Handwriting

handwritten

63 pages
394 lines
10,261 chars

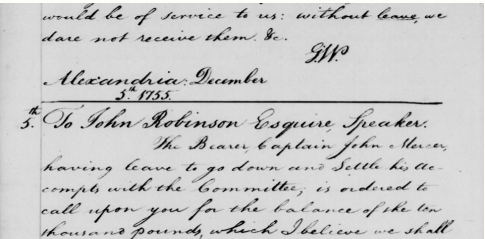


IAM

English Modern Handwriting

handwritten

20 pages
180 lines
7,604 chars



Washington

Historical English Manuscript

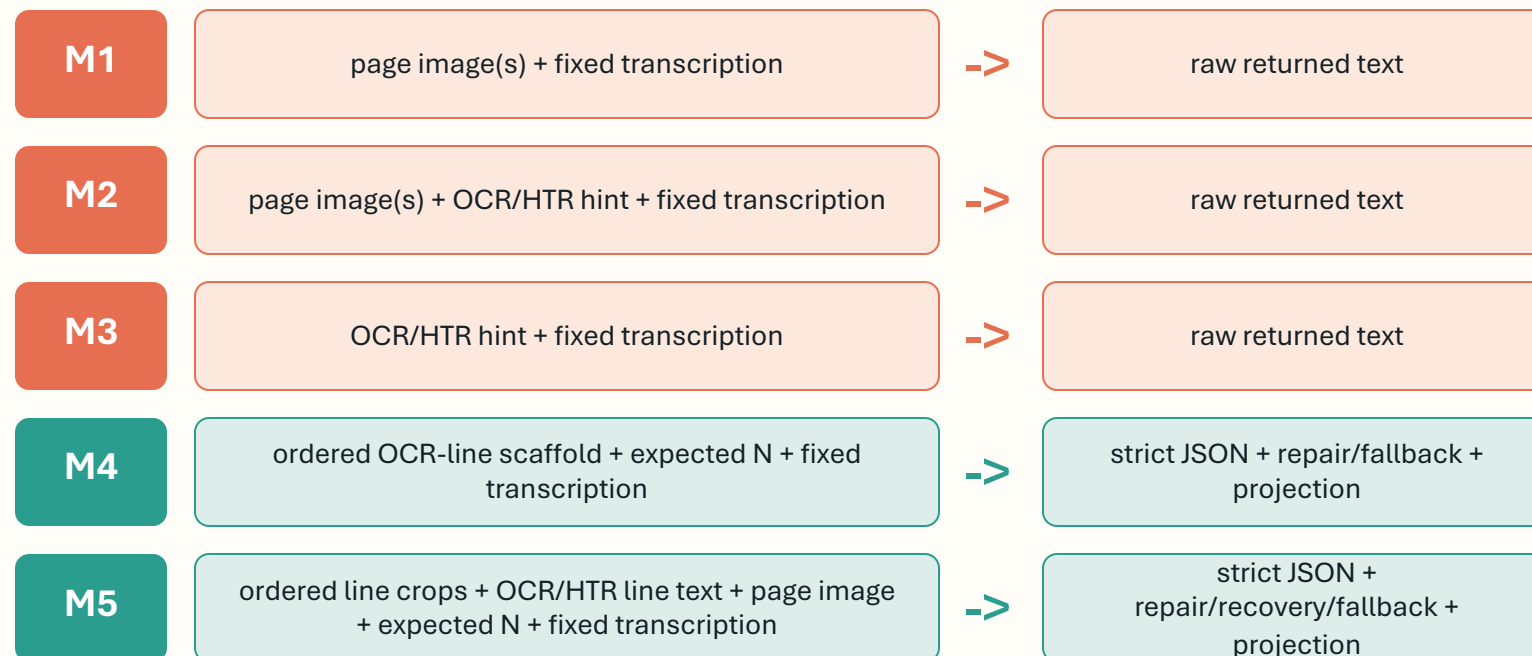
handwritten

20 pages
656 lines
26,458 chars

Printed

Bullinger Print: 10 samples/pages, 271 lines, 15,004 chars
IAM Print: 30 pages/forms, 161 lines, 12,194 chars

LLM/VLM Setups



NNTP Reference

Line images + recognizer evidence + fixed transcription

→ Deterministic forced alignment

Separate Non-Prompt Reference

Models: GPT-5.4-0305, GPT-5.2, Gemini-2.5-Pro, Gemini-3-Pro, Gemini-3.1 Qwen3-VL-8B, Qwen3-VL-32B, Mistral-2512

What exactly is calculated?

Same character stream; one missing newline merges Alpha and Beta.

Metric Example

Ground truth

- 1 Alpha
- 2 Beta
- 3 Gamma
- 4 Delta

Prediction

- 1 Alpha Beta
- 2 Gamma
- 3 Delta

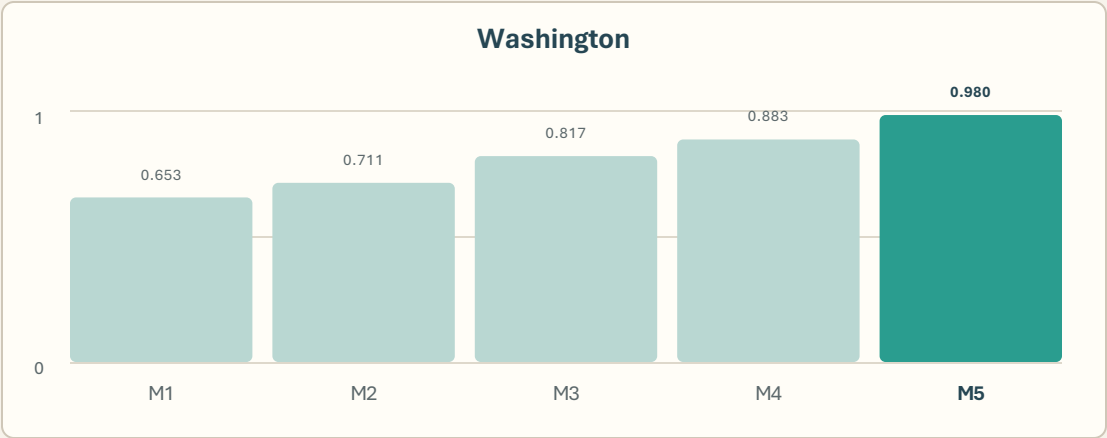
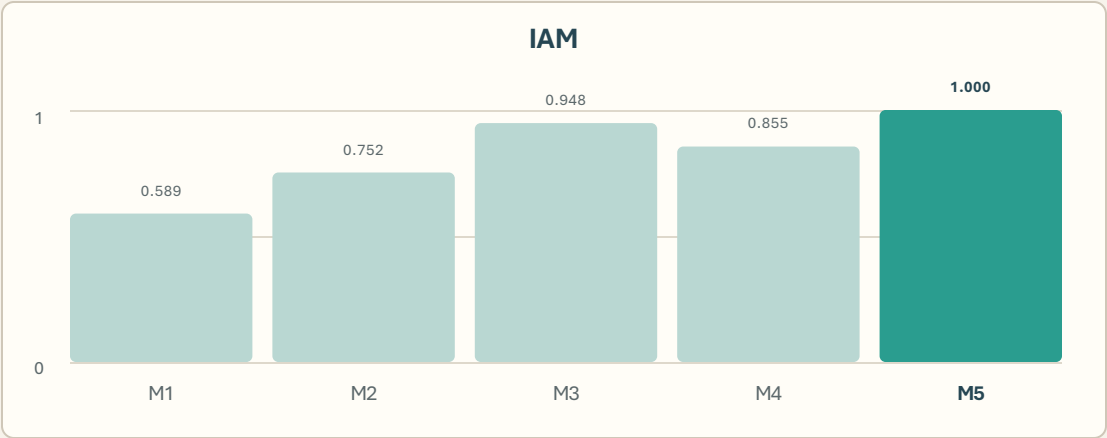
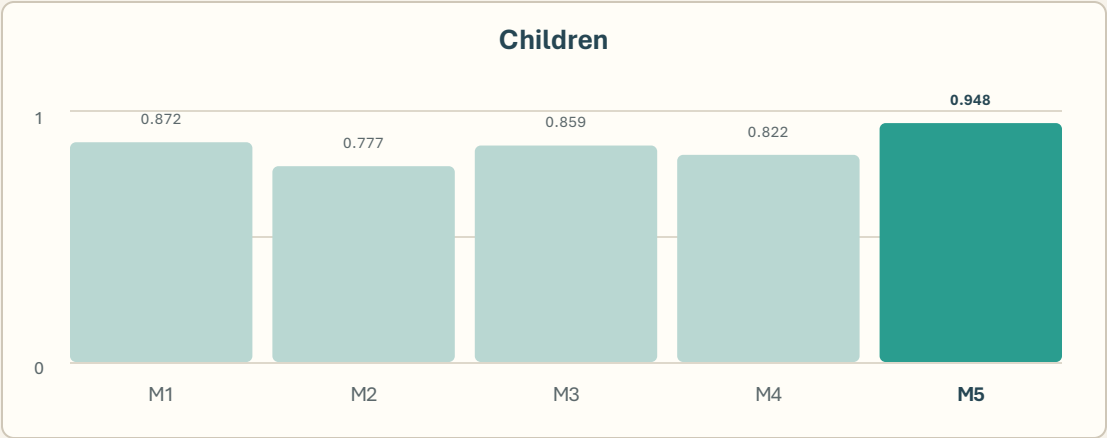
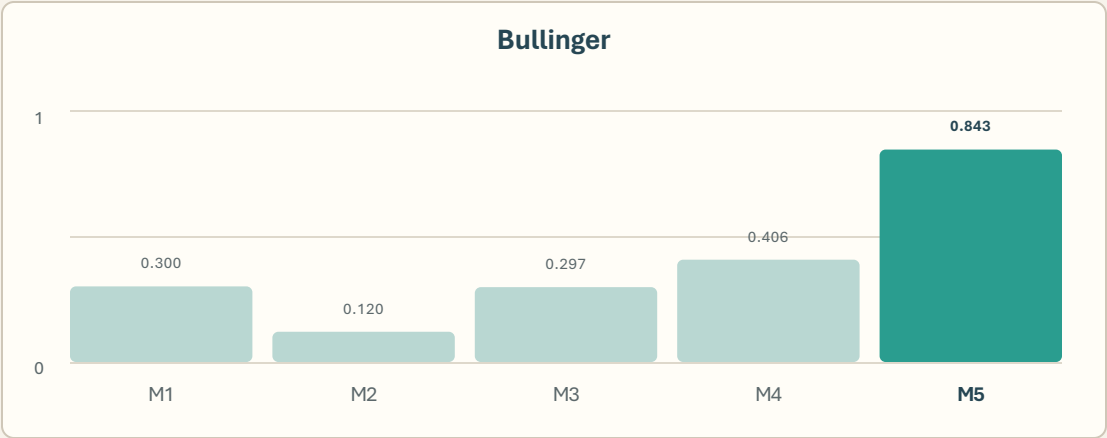
After whitespace normalization, both full texts are:
Alpha Beta Gamma Delta

$n = \max(4, 3) = 4$; exact matched lines: Gamma, Delta.

Metric formulas and plug-in values

Metric	Formula / calculation	Result
LA->	top-down exact line matches / n	0 / 4 = 0.000
LA<-	bottom-anchored exact line matches / n	2 / 4 = 0.500
LAnorm+/-	(LA->norm + LA<-norm) / 2	(0 + 0.500) / 2 = 0.250
F1,line	$P=2/3, R=2/4; F1 = 2PR / (P + R)$	0.571
CERnorm	$\text{lev}(\text{chars}(Nx), \text{chars}(Ny)) / \text{chars}(Nx) $	0 / 22 = 0.000
WERnorm	$\text{lev}(\text{words}(Nx), \text{words}(Ny)) / \text{words}(Nx) $	0 / 4 = 0.000

CER/WER see identical normalized text, while line metrics expose the lost boundary between Alpha and Beta.



Metric: LAnorm±

Intermediate methods are dataset-dependent; no monotonic component ablation is claimed.

M5 Is Highest-Scoring Among Evaluated LLM/VLM Methods.

Results Per Model And Dataset

Whitespace-normalized LAnorm+/- macro scores. Means are unweighted over represented cells; partial rows are context, and dashes stay missing/ineligible.

Complete four-dataset						
Configuration	Cov	Mean	B	C	IAM	W
GPT-5.4-0305 / M5	4/4	0.940	0.843	0.948	1.000	0.970
Gemini-2.5-Pro / M5	4/4	0.786	0.423	0.846	0.992	0.883
GPT-5.4-0305 / M3	4/4	0.730	0.297	0.859	0.948	0.817
GPT-5.4-0305 / M4	4/4	0.705	0.406	0.822	0.855	0.737
Mistral-2512 / M4	4/4	0.610	0.380	0.351	0.830	0.876
GPT-5.4-0305 / M2	4/4	0.590	0.120	0.777	0.752	0.711
GPT-5.4-0305 / M1	4/4	0.554	0.300	0.804	0.458	0.653
Mistral-2512 / M3	4/4	0.405	0.067	0.556	0.789	0.208
Gemini-3.1 / M1	4/4	0.397	0.062	0.872	0.589	0.065
Qwen3-VL-8B / M2	4/4	0.234	0.040	0.352	0.000	0.545
Mistral-2512 / M2	4/4	0.234	0.027	0.401	0.246	0.263
Qwen3-VL-8B / M1	4/4	0.225	0.108	0.485	0.003	0.305
Mistral-2512 / M1	4/4	0.216	0.046	0.525	0.111	0.181
Qwen3-VL-8B / M3	4/4	0.173	0.002	0.190	0.424	0.076

Partial coverage						
Configuration	Cov	Mean	B	C	IAM	W
NNTP / NNTP	3/4	0.976	-	0.939	1.000	0.990
Qwen3-VL-32B / M5	3/4	0.962	-	0.923	0.982	0.980
Mistral-2512 / M5	3/4	0.911	-	0.846	0.930	0.957
Qwen3-VL-8B / M4	3/4	0.672	0.398	-	0.736	0.883
Gemini-2.5-Pro / M2	2/4	0.873	-	0.868	-	0.879
Gemini-2.5-Pro / M1	2/4	0.867	-	0.858	-	0.876
GPT-5.2 / M1	2/4	0.803	-	0.846	-	0.761
GPT-5.2 / M2	2/4	0.795	-	0.837	-	0.752
Gemini-2.5-Pro / M3	2/4	0.286	-	0.095	-	0.477
GPT-5.2 / M3	2/4	0.226	-	0.066	-	0.386
Gemini-3-Pro / M2	2/4	0.201	-	0.377	-	0.024
Gemini-3-Pro / M1	2/4	0.190	-	0.360	-	0.019
Gemini-3-Pro / M3	2/4	0.044	-	0.066	-	0.022
Gemini-3.1 / M2	1/4	0.092	-	-	-	0.092
Gemini-3.1 / M3	1/4	0.073	-	-	-	0.073

Fixed-Transcription Preservation

Strict JSON

Repair If Parse/Count Fails

Recover Or Fallback If No Plan

Project Boundaries

Score Re-Sliced Fixed Text

Projection Mechanism

Model / JSON Line Hint

"The Bearer, Caplain"
"John Mercer"



Fixed Transcription

"The Bearer, Captain John Mercer"



Projected Output

"The Bearer, Captain
John Mercer"

Boundary From Hint; Characters From Fixed Text.

Validity Evidence

0 / 162 M5 preservation failures.

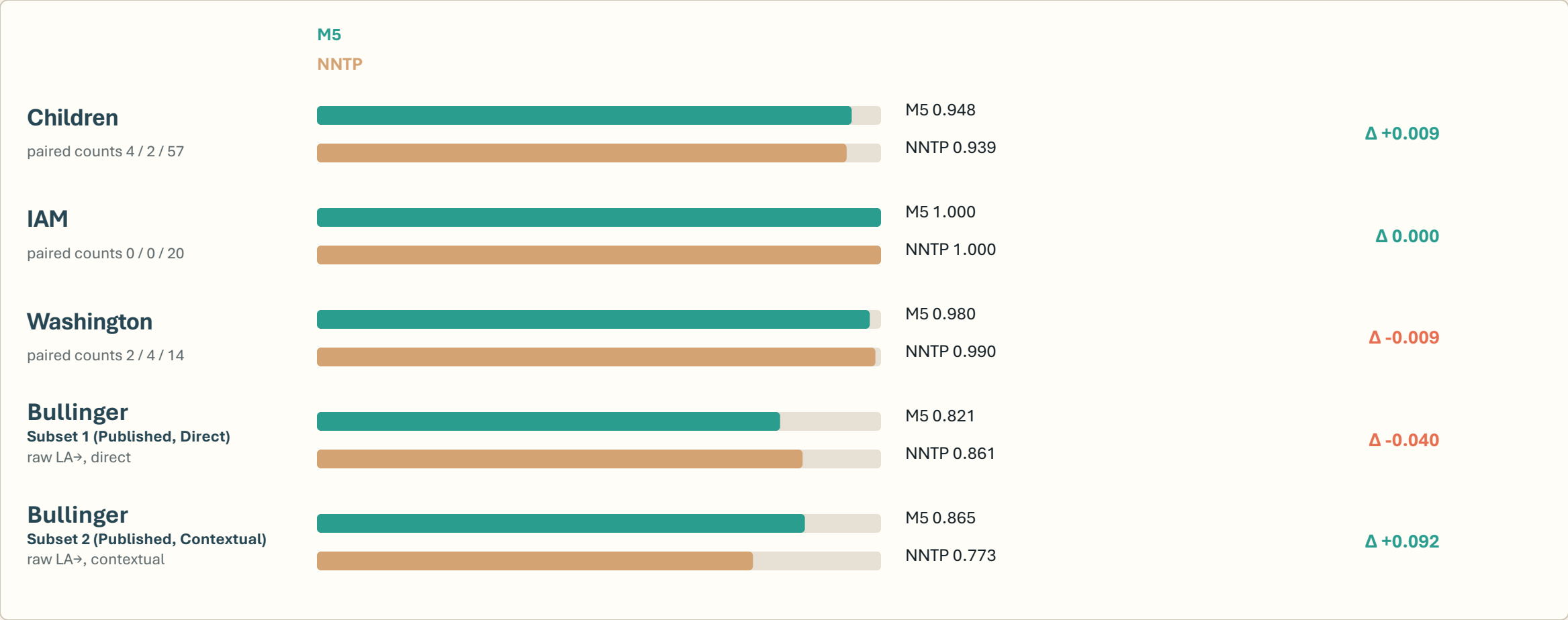
M1-M3 Average line-count drift: 8.195

Raw generations often returned too many or too few lines.

M4-M5 Average line-count drift: 0.059

Structured validation/projection almost eliminates line-count drift.

Known-count staging: M4/M5 validate against stored line counts.



Washington 304

Ground truth

1 | 304. Letters Orders and Instructions. December 1755.
2 | be sure of idemnity. If your Honor would
3 | be kind enough to intimate this to General
4 | Shirley, or the Colonels of those Regiments,- it
5 | would be of service to us: without leave we

M3 prediction

CERnorm 0.000
LAnorm+/- 0.338

1 | 304. Letters Orders and Instructions. December 1755. be
2 | sure of idemnity. If your Honor would be kind enough
3 | to intimate this to General Shirley, or the Colonels
4 | of those Regiments,- it would be of service to us:
5 | without leave we dare not receive them. c. GW

M3: 'be' joins
header line

Boundary after the
header is shifted.

M5 prediction

CERnorm 0.000
LAnorm+/- 1.000

1 | 304. Letters Orders and Instructions. December 1755.
2 | be sure of idemnity. If your Honor would
3 | be kind enough to intimate this to General
4 | Shirley, or the Colonels of those Regiments,- it
5 | would be of service to us: without leave we

M5: break after
December 1755.

Established Within Scope

- Boundary prediction over fixed text
- M5 is highest-scoring among evaluated LLM/VLM setups
- Output control is central

Limits

- Close and dataset-dependent vs. NNTP
- Known-count structured setting
- Configuration-Level Evidence

Future Work

- Blind line-count / segmentation setting
- Component-level isolation
- Direct boundary prediction / constrained decoding

From Prompting To Controlled Alignment

Newline-only alignment moves from weakly constrained prompting to structured boundary assignment.

1

VLMs can use line-level visual evidence for transcription alignment.

2

Output control is what makes the result valid under fixed transcription.

3

The highest-scoring evaluated VLM setup is competitive with NNTP.

Thank you!

Given fixed transcription x , predict y such that $\text{remove_newlines}(y) = x$.

$Y(x)$ = all strings formed by inserting zero or more newline characters between existing characters of x

Boundary-vector view: choose admissible insertion positions; do not change the underlying character sequence.

Dataset Inventory

Dataset	Modality	n	Pages	Lines	Chars
Bullinger Handwritten	handwritten	59	100	2,535	133,883
Children Handwritten	handwritten	63	63	394	10,261
IAM Handwritten	handwritten	20	20	180	7,604
Washington Handwritten	handwritten	20	20	656	26,458
Bullinger Print	printed	10	10	271	15,004
IAM Print	printed	30	30	161	12,194

Method	Input evidence	Output control	Scored output
M1	page image(s) + fixed transcription	prompt only	raw returned text
M2	page image(s) + OCR/HTR hint + fixed transcription	prompt only	raw returned text
M3	OCR/HTR hint + fixed transcription	prompt only	raw returned text
M4	ordered OCR-line scaffold + expected N + fixed transcription	strict JSON, repair/fallback, projection	projected boundary plan
M5	ordered line crops + OCR/HTR line text + page context + expected N + fixed transcription	strict JSON, repair/recovery/fallback, projection	projected boundary plan
NNTP	line images + recognizer output + fixed transcription	deterministic forced alignment; PyLaia filtering	filtered reconstruction

Typical Failure Modes By Method

Method	Typical failure mode	Why it happens	Example / evidence	Interpretation
M1	Raw Copy + Segment Drift	Copying and boundary inference happen in one raw generation; no projection.	452/648 non-space drift; 295 under, 78 over. Bullinger 11084: missing opener; LAnorm+/- 0.484, F1norm 0.984.	Image evidence can help, but fixed-text validity is not guaranteed.
M2	Noisy OCR Hint Followed	Image, fixed text, OCR/HTR hint, and chunks must be reconciled.	M2-M1 worse in 233/506. Children 2A_11_18-19: M1 exact; M2 one line (Qwen3-VL-8B).	OCR/HTR is a noisy structural hint, not monotonic extra information.
M3	No Visual Correction	OCR/HTR hint plus fixed text only; raw returned text is scored.	271/506 under-segmented. Washington 304: CERnorm 0.000, LAnorm+/- 0.338.	Near-ceiling when lineation is good; limited when the scaffold is wrong.
M4	Flawed OCR Scaffold	Exact count and JSON/projection can still follow bad OCR splits.	Washington 307: M3 exact; M4 LAnorm+/- 0.000 despite F1norm 0.839.	Validity control improves form, not guaranteed boundary accuracy.
M5	Hard Local Boundary / Fallback Context	Line crops, OCR text, page context, repair/fallback, and projection remain entangled.	17 hyphen mismatches; fallback LAnorm+/- 0.573 vs projection 0.920. Washington 270: M5 0.871 < M3 0.935.	Highest-scoring evaluated setup, not an every-sample winner.

Illustrative mechanisms from stored prediction/trace analysis; not causal component ablations.

M1 Failure Example: Missing Opening Line

M1

Raw Copy + Segment Drift

Example 1/5
Stored prediction excerpt

Bullinger 11084 · GPT-5.4-0305

Ground truth opening

L01: 20154
L02: Op D. Lieber gunstiger Heu vnd bruder, das ich nun schriben an
Ich
L03: So hang verzogen hab ist kein andere Vrsach dan das ich alwege
verhoffett hat
L04: Hans schmid wurde hinuber komen, das ich inne brief durch ein
gewisser

Actual M1 prediction opening

L01: Op D. Lieber gunstiger Heu vnd bruder, das ich nun schriben an
Ich
L02: So hang verzogen hab ist kein andere Vrsach dan das ich alwege
verhoffett hat
L03: Hans schmid wurde hinuber komen, das ich inne brief durch ein
gewisser
L04: botte vch vberantwurten mocht das aber mit bestheilen etc. vff
Sommlichen

Prediction starts at the second ground-truth line; the opening line “20154” is absent.

LAnorm+/- 0; reverse LAnorm 0.969; F1norm 0.984; CERnorm 0.003.

Short verbatim windows from retained predictions; examples illustrate mechanisms, not causal component ablations.

M2 Failure Example: OCR Hint Followed Too Coarsely

M2

Noisy OCR Hint Followed

Example 2/5
Stored prediction excerpt

Children 2A_11_18-19 · Qwen3-VL-8B

Ground truth has seven lines

L01: Ein grosses blaues waser
L02: sit man. Der leuchttarm Jeuchtet.
L03: Man siet auch eine insel.
L04: Ein Kreps & pakt den man
L05: an den Oren und der man
L06: sagt aaa aua und weint
L07: fast un das so weetut.

Actual M2 prediction collapses to one line

L01: Ein grosses blaues waser sit man. Der leuchttarm Jeuchtet. Man
siet auch eine insel. Ein Kreps & pakt den man an den Oren und der
man sagt aaa aua und weint fast un das so weetut.

The same model is exact in M1, but M2 returns the whole text as one metric line.

M1 LAnorm+/- 1 vs M2 LAnorm+/- 0; M2 lines pred/gt 1/7.

Short verbatim windows from retained predictions; examples illustrate mechanisms, not causal component ablations.

M3 Failure Example: No Visual Correction

M3

OCR/HTR-Only Under-Segmentation

Example 3/5
Stored prediction excerpt

Washington 304 · GPT-5.4-0305

Ground truth opening

L01: 304. Letters Orders and Instructions. December 1755.
L02: be sure of idemnity. If your Honor would
L03: be kind enough to intimate this to General
L04: Shirley, or the Colonels of those Regiments,- it
L05: would be of service to us: without leave we
L06: dare not receive them. c.

Actual M3 prediction opening

L01: 304. Letters Orders and Instructions. December 1755. be
L02: sure of idemnity. If your Honor would be kind enough
L03: to intimate this to General Shirley, or the Colonels
L04: of those Regiments,- it would be of service to us:
L05: without leave we dare not receive them. c. GW
L06: Alexandria: December 5th. 1755.

The text stream normalizes correctly, but opening boundaries are merged and shifted.

CERnorm 0; LAnorm+/- 0; F1norm 0.708; 31/34 predicted lines.

Short verbatim windows from retained predictions; examples illustrate mechanisms, not causal component ablations.

M4 Failure Example: Correct Count, Wrong Boundary

M4

Flawed OCR Scaffold

Example 4/5
Stored prediction excerpt

Washington 307 · GPT-5.4-0305

Ground truth header/date

L01: Letters Orders and Instructions. December 1755.
L02: 10th. Alexandria December 10th. 1755.
L03: A Sergeant and twelve men, to be
L04: in readiness to escort a waggon up to Winchester to-
L05: morrow.

Actual M4 projected prediction

L01: Letters Orders and Instructions.
L02: December 1755.
L03: 10th. Alexandria December 10th. 1755.
L04: A Sergeant and twelve men, to be
L05: in readiness to escort a waggon up to Winchester to-
L06: morrow.

M4 keeps the expected line count, but splits the opening header/date and shifts the sequence.

M3 exact; M4 LAnorm+/- 0 despite F1norm 0.839; M4 lines pred/gt 31/31.

Short verbatim windows from retained predictions; examples illustrate mechanisms, not causal component ablations.

M5 Failure Example: Local Hyphen Boundary Shift

M5

Hard Local Boundary / Repair Context

Example 5/5
Stored prediction excerpt

Washington 270 · Qwen3-VL-32B

Ground truth hyphenated continuations

L14: The Officers who came down
L15: from Fort Cumberland with Colonel
L16: Washington, are immediately to go Recrui-
L17: ting; and they are allowed until the 1st. of De-
L22: They are to wait upon the Aid de camp
L23: at one of the Clock, to receive their Recrui-
L24: ting Instructions. Each Officer present, to give

Actual M5 projected prediction

L14: The Officers who came down
L15: from Fort Cumberland with Colonel
L16: Washington, are immediately to go Recrui
L17: - ting; and they are allowed until the 1st. of De-
L22: They are to wait upon the Aid de camp
L23: at one of the Clock, to receive their Recrui
L24: - ting Instructions. Each Officer present, to give

The local split moves the hyphen from the end of “Recrui-” to the start of the next line.

M5 LAnorm+/- 0.871 vs M3 0.935; trace attempts: initial+repair; repair trigger “expected 31 lines but got 29”.

Short verbatim windows from retained predictions; examples illustrate mechanisms, not causal component ablations.

LAnorm+/-

primary

Average of forward and reverse order-sensitive line accuracy after whitespace normalization.

F1_line norm

supporting

Order-insensitive multiset exact-line matching; useful when many line strings are present but shifted.

CERnorm / WERnorm

supporting

Sanity checks for residual character or word drift after full-text whitespace normalization.

Diagnostic matched-support comparison: whitespace-normalized LAnorm+/-

Dataset	n	M1 HW	M1 Print	M2 HW	M2 Print	M3 HW	M3 Print
support slice		LAnorm+/-	LAnorm+/-	LAnorm+/-	LAnorm+/-	LAnorm+/-	LAnorm+/-
Bullinger	10	0.148	0.594	0.083	0.566	0.066	0.115
IAM	30	0.959	0.917	0.912	0.896	0.802	0.934

Printed scope remains M1-M3 only; no printed M4, M5, or NNTP rows were run.

Model	M1	M2	M3
GPT-5.2	-0.042	-0.122	+0.039
Qwen3-VL-8B	+0.037	+0.027	+0.078

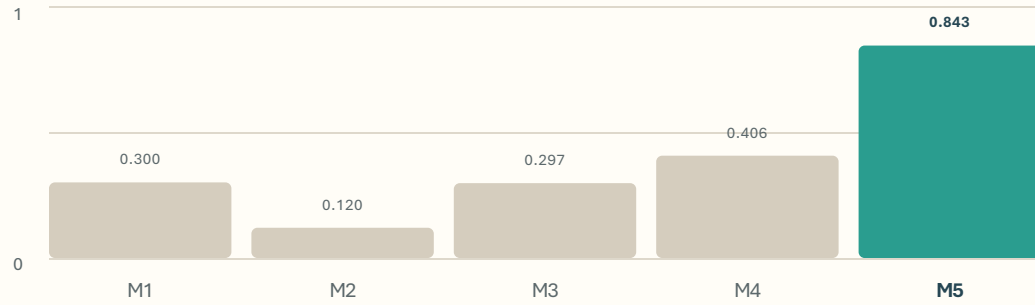
Exploratory early check: one example can help, do nothing, or hurt depending on model and method.

Subset	Thesis M5 raw LA->	Published NNTP raw LA->	Interpretation
Subset 1	0.821	0.861	direct; published NNTP higher
Subset 2	0.865	0.773	contextual; inventories differ

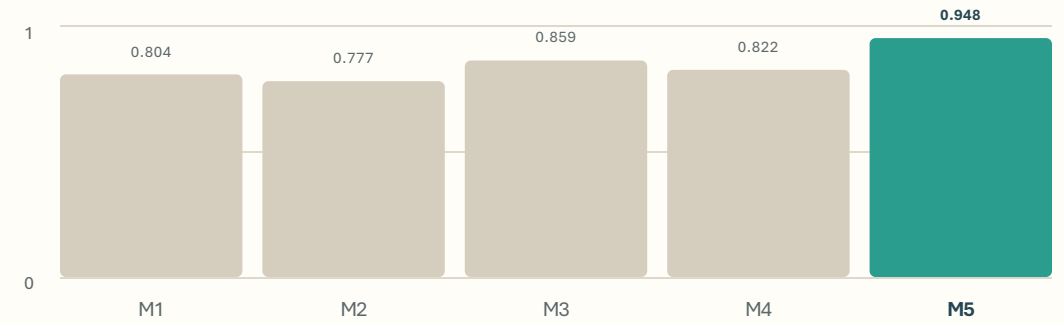
Not a full-dataset M5-vs-NNTP winner claim.

Exact OpenAI Control

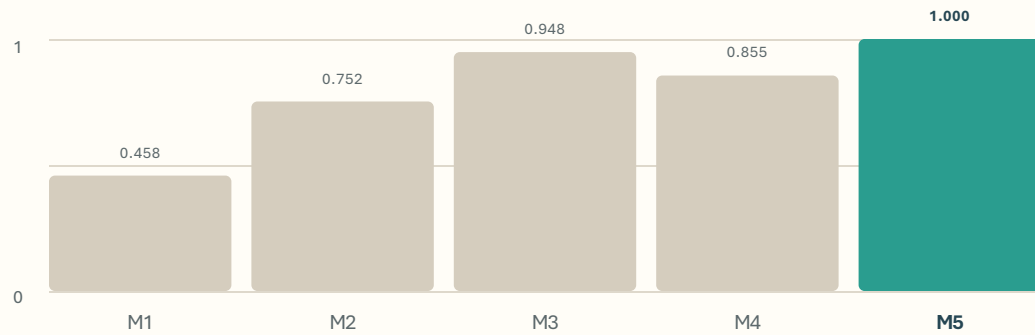
Bullinger



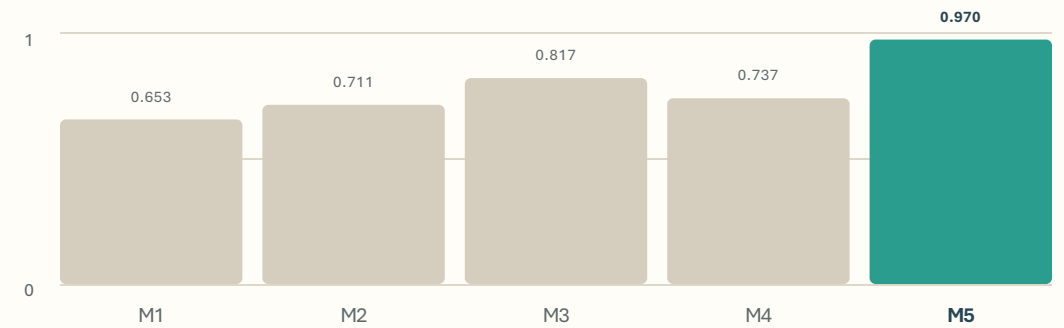
Children



IAM



Washington



Metric: LAnorm±

Backup Evidence: Single OpenAI Configuration; M5 Is Highest-Scoring In Each Dataset.

Dataset	Samples	1st JSON	Repair	Recovery	Fallback	Final proj.	Fail	Missing
Bullinger	59	54	2	3	3	39	0	0
Children	63	63	0	0	0	58	0	0
IAM	20	20	0	0	0	20	0	0
Washington	20	20	0	0	0	19	0	0

157/162

first JSON exact-
count

Validity Boundaries

known-count staging for M4/M5

model-configuration dependence

M1/M2 multi-page chunking

small or mismatched sample sets

mixed line-crop provenance

projection vs raw returned text

NNTP PyLaia symbol filtering

missing deterministic scaffold ablations

CTC loss

image time steps
+ char/blank probabilities

>

many paths collapse
to the same target text

>

loss = -log sum
of valid paths

No manual character positions are needed.
Blanks and repeats let many paths explain the same line text.

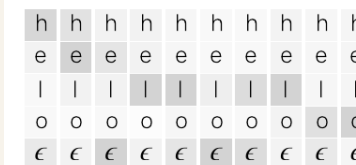
or an output sequence.



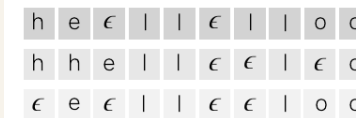
We start with an input sequence, like a spectrogram of audio.



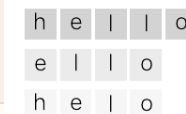
The input is fed into an RNN, for example.



The network gives $p_t(a | X)$, a distribution over the outputs $\{h, e, l, o, \epsilon\}$ for each input step.



With the per time-step output distribution, we compute the probability of different sequences



By marginalizing over alignments, we get a distribution over outputs

Hannun, Distill 2017, CC-BY 4.0

NNTP: how line breaks are recovered

ordered line crops
+ CTC observations

>

fixed transcription
characters unchanged

>

token passing /
forced alignment

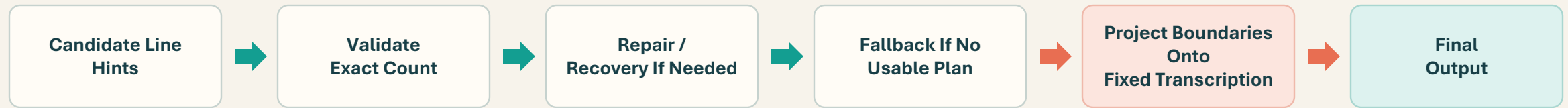
>

newlines at
line-image breaks

NNTP uses CTC observations at inference

PyLaia syms.txt caveat: unsupported characters can be stripped before NNTP alignment.

How Fixed Transcription Is Preserved



Captain -> Captain Projection Example

Model Hint:

"The Bearer, Caplain "
"John Mercer"

Fixed Transcription:

"The Bearer, Captain John Mercer"

Projected Output:

1 | The Bearer, Captain
2 | John Mercer

Projection Uses Cumulative Hint Offsets Mapped To The Fixed Transcription.

Q&A Takeaways

- Model Text Is Boundary Evidence, Not Final Content.
- Projection Re-Slices The Untouched Transcription.
- Successful Projection Preserves The Non-Newline Character Stream.
- Known-Count Staging: M4/M5 Use Stored Expected Line Count.

Thesis Anchors: Section 5.7, Appendix A.3, Table 6.7, Table 6.8

- blind line-count / segmentation setting
- deterministic scaffold references
- ablations of M5 components
- dedicated boundary prediction
- constrained decoding instead of post-hoc projection
- broader datasets and stronger crop pipelines

Core direction

predict boundaries directly over the fixed
transcription

M5 vs NNTP: concrete non-tie cases

Tie-dominated: Children 57/63, IAM 20/20, Washington 14/20

Dataset	Sample ID	Higher LAnorm	Delta LAnorm	Observed mechanism	Local excerpt evidence	Evidence file
Children	2A_12_8-9	M5	+1.000	NNTP merged first two lines and split a short final fragment.	GT/M5: "...Ka&Ze und ain Hund Der Ma" NNTP: "...Ka&Ze und ain Hund Der Ma"	M5 trace JSON NNTP pred TXT
Children	2A_12_10-11	NNTP	-0.500	M5 shifted "Saine" to the previous line; NNTP keeps the local boundary.	GT/NNTP: "Der Man TuT Saine Claider..." M5: "Der Man TuT Saine Claider..."	M5 trace JSON NNTP pred TXT
Washington	278	M5	+0.063	NNTP split short "andria." and merged the remainder into the next numbered line.	GT/M5: "Alex- andria. 1st. To..." NNTP: "Alex- andri a. 1st. To..."	M5 trace JSON NNTP pred TXT
Washington	272	NNTP	-0.059	M5 merged a header/date boundary; NNTP matches the date split.	GT/NNTP: "...Instructions. October 28th. To..." M5: "...Instructions. October 28th. To..."	M5 trace JSON NNTP pred TXT

Observed mechanism

- Non-ties are local boundary placements.
- Repeated sites: short fragments, header/date lines, hyphenated continuations, and mid-word splits.
- Mixed cases are not shown here.

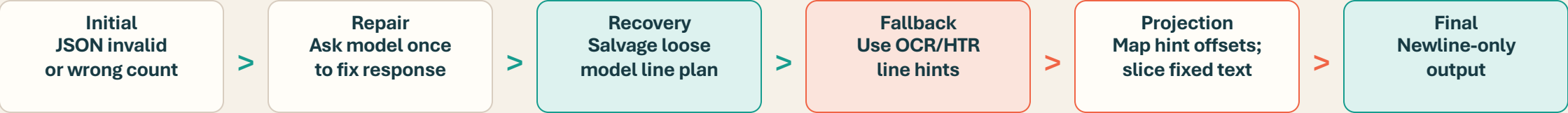
Possible explanation supported by artifacts

- M5 records line crops, page context, OCR/HTR hints, expected line count, and projection/fallback paths.
- NNTP records recognizer-derived forced-alignment output.
- No ablation proves a stronger causal story.

How Fallback And Recovery Work

Safety path when strict M5 JSON or expected line count validation fails.

Boundary hints only: final characters come from fixed transcription.



Recovery is tried before fallback: the system prefers a usable model boundary plan, but falls back to deterministic line hints if no plan can be reconciled.

Tiny recovery example

Fixed transcription: abcdefgh
Expected lines: 4
Loose model plan: ab | cdef | gh (wrong count)
OCR/HTR hint: ab | cd | ef | gh
Recovered hints: ab | cd | ef | gh
Projected final: ab | cd | ef | gh

The recovered text is not copied as final content. It supplies boundary positions.

Definitions

- Repair: ask once for valid JSON and count.
- Recovery: reconcile loose model lines with expected count and OCR/HTR hints.
- Fallback: use OCR/HTR line texts as boundary hints when model hints fail.
- Projection: map hint offsets, then slice fixed text.

Not OCR/HTR replacement; boundary projection only.

Source: M5 inference code and regression tests; mechanism shown as implemented behavior, not a metric claim.